Forced and Free Content in the GeoVac Framework: An Observations Paper

J. Loutey

June 2026

Abstract

The GeoVac framework autonomously generates a wide class of structural facts (graph spectrum, S^3 outer manifold, gauge group $U(1) \times SU(2) \times SU(3)$, master Mellin engine, inner-algebra factor count, the matter-sector moduli dimension $\dim \mathcal{M}(D_F)\big|_{\text{matter}} = 128$ per generation (full-axiom dimension 260), two-term-exact gravity, and many more) and consumes specific numerical content as external input (Yukawa values, generation count, calibration cutoff moments, atomic-physics screening exponents, multi-loop QED counterterms, etc.). We catalogue both sides of this *forced/free seam* across the corpus — 35 forced entries, 1 admitted-not-forced entry, 24 calibration entries, spanning foundations, gauge structure, periods, gravity, QED, the Standard Model inner factor, multi-focal-composition walls, chemistry / quantum computing scaling, and foundational calibration. We identify a cross-domain unity pattern (the same wall appears in six independent settings, all carrying the same multi-focal-composition signature) and test five candidate principles for what discriminates the two sides: multi-focal depth, period-content classifiability, dimensional character, compactness inheritance, and packing-reachability. The fifth candidate (packing-reachability: an entry is forced iff a witness structural derivation exists from {Paper 0 + standard real-spectral-triple axioms + Hopf-tower truncation + Bertrand + Upgrade B + CCM canonical SM representation}) *consolidates* the forced/free boundary rather than independently discriminating it: its 98.3% agreement with the F/C status (59/60) is an internal-consistency check confirming the catalogue applies its conventions uniformly — the packing-reachable tag is the F/C label by construction — not an independent accuracy. The independent discriminator signal is carried by multi-focal depth, period class, and the two-family decomposition. The earlier-named two-family decomposition (multi-focal composition vs. inner-factor input data) is preserved as the structural failure-mode story beneath the principle: both families share the same surface verdict (packing-unreachability) but exhibit different structural reasons for it. This paper consolidates what was previously scattered across CLAUDE.md §1.7 working hypotheses, dead-end records, and individual sprint memos, and adds the Sprint C2 principle hunt that closes the question Paper 57 §6.1 originally left open.

1 Introduction: the seam and why naming it matters

GeoVac is a discretization framework built on the Paper 0 packing axiom [1] and its consequence (proved in Paper 7) that the resulting graph Laplacian on the unit three-sphere S^3 is conformally equivalent to the continuous Schrödinger problem with $-Z/r$ potential [2]. From this minimal foundation the framework has been shown to autonomously generate a large class of structural facts: the universal kinetic scale $\kappa = -1/16$, the angular sparsity theorem of Paper 22 [5], the chemistry $O(Q^{2.5})$ Pauli count scaling of Paper 14 [3], the spectral triple structure of Paper 32 [9] with proven Gromov–Hausdorff convergence to the continuous S^3 spectral triple at rate $4/\pi$ (Paper 38 [13]), the gauge group $U(1) \times SU(2) \times SU(3)$ as a structural consequence of Bertrand $\times$ Hopf-tower truncation (Paper 32 §VIII.B), the master Mellin engine classification of every transcendental appearing in a GeoVac observable (Paper 18 §III.7, Paper 32

§VIII case-exhaustion theorem [4, 9]), and the Forced-Count Theorem (matter-sector moduli dimension $\dim \mathcal{M}(D_F)\big|_{\text{matter}} = 128$ per generation; full-axiom dimension 260) (Paper 32 §VIII, `thm:forced_count`).

The framework also consumes specific numerical content that it does not autonomously generate. The Yukawa values cannot be selected by the inner-factor structure (Sprint H1 non-selection theorem, empirically confirmed at 162 PSLQ cells against the inner-factor period ring $M_1 \cup M_2$ [9, 23]). The generation count $N_{\text{gen}} = 3$ is packing-unreachable in the current axiom system (Direction 2 NO-GO, Read 2 NO-GO [25]). The chemistry-side inner-region correlation wall (*W1e*) consumes external atomic-physics screening data (Clementi–Raimondi exponents) and cannot be closed by any framework-internal rational/algebraic modification, as verified by six independent sprint attempts (CLAUDE.md §3 entries F1–F6 plus Schmidt, core correlation, and basis enlargement). The multi-loop QED renormalization counterterms (Z_2, δm) are likewise consumed rather than generated (LS-8a wall [12]).

The line between what the framework forces and what it consumes is itself a structural property of the framework. It has been repeatedly named — the *structural-skeleton scope* reading of CLAUDE.md §1.7 WH3, the *multi-focal-composition wall pattern* crystallized at six instances, the *external-input three-class partition* (Calibration / Multi-focal composition / Multi-determinant correlation) — but never consolidated. Each naming has produced a clearer statement than the previous one, but the underlying inventory of which physical inputs sit on each side, and which discriminator (if any) governs the partition, has been distributed across many sprint memos, dead-end rows, and working-hypothesis bullets.

This paper consolidates the inventory and tests, at preliminary level, four candidate discriminator principles against it. The goal is not to close the seam — we do not yet claim to have identified *the* principle that decides which side a given physical input falls on — but to name the question sharply enough that it admits a determinate answer.

Why an observations paper, not a theorem paper. The format follows Paper 2, Paper 25, and Paper 33 (the existing observation-genre papers in the GeoVac corpus). We describe a structural pattern that the framework exhibits across multiple independent domains, we cite the underlying theorems and dead-end records that establish each entry, and we probe candidate discriminators without commitment. Where a finding has theorem-grade status it is named as such; where it is a numerical / empirical observation only, the honest scope is recorded explicitly. The companion catalogue `docs/forced_free_seam.md` carries the full table; this paper is the publication-grade framing.

What we promise to deliver in this paper.

- §2 — forced-side inventory across eight domains (foundations, gauge, periods, gravity, QED, Standard Model inner factor, chemistry/QC scaling, foundational calibration; the load-bearing sub-domains).
- §3 — free-side inventory with non-selection theorems and documented negative searches.
- §4 — the cross-domain unity: the multi-focal-composition wall pattern at six independent instances.
- §5 — formal probe of five candidate discriminators (Sprint C2 finding: packing-reachability is the consolidated F/C boundary; its 98.3% agreement with the F/C status is an internal-consistency check, not an independent discriminator).
- §6 — open questions; the "second packing axiom" status.

What we do NOT promise.

- A theorem stating "GeoVac forces X iff X has property P." No such theorem is proved or even decisively favoured in this paper.
- A derivation of any presently-free quantity from a single GeoVac principle.
- A demonstration that the present seam is the only possible seam (a "second packing axiom" could shift the line and remains the named long-range frontier).

2 The forced side

We organize the forced inventory by sub-domain. Each entry carries a mechanism (what does the forcing), a witness (theorem cite or bit-exact verification), and three discriminator columns: multi-focal depth (MF), period class in the master Mellin engine (P; values M_1, M_2, M_3, "outside," "none," "meta" for the engine itself, "N/A" for non-period observables), and dimensional character (Dim; dimensionless ratio vs. dimensionful scale). Sub-domains are referenced by the letter codes A–I used in the companion catalogue `docs/forced_free_seam.md`.

2.1 Foundations / skeleton (A)

The packing axiom of Paper 0 outputs a $\mathbb{Z}$-graded combinatorial structure whose first concrete physical content is the spectrum $\lambda_n = n^2 - 1$ of the continuum S^3 Laplace–Beltrami operator that the (positive-semidefinite) graph Laplacian converges to [1, 2]. Paper 7's eighteen symbolic proofs of S^3 conformal equivalence [2] pin the outer manifold; Bertrand's classical theorem [18] combined with SO(4) closure forces it to be S^3 rather than a different homogeneous space. The universal kinetic scale $\kappa = -1/16$ is the chordal-distance projection coefficient, exact rational. The shell degeneracy $g_n = n^2$ is representation-theoretic. The deeper organizing observation, established across Papers 18, 22, 24, 27, 32, and consolidated in the structural-skeleton scope pattern, is that the framework's discrete content is exactly the closed/compact/Peter–Weyl regime; the continuum is what remains when compactness is released. (*Discreteness* $\leftrightarrow$ *compactness*, A5.)

2.2 Gauge structure (B)

The Standard Model gauge group emerges from two independent structural inputs: Bertrand's classical theorem (only $-Z/r$ and kr^2 central potentials produce closed orbits) and the Hopf-tower of spheres $S^1 \to S^3 \to S^5 \to \cdots$ truncated at $n \le 3$ by the Hurwitz classification of associative real normed division algebras (Door 4d, summarized in [9], §VIII.B). This produces

$$\mathcal{A}_F \;=\; \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \tag{1}$$

with $\mathbb{C}$ at $n = 1$ from the unique division algebra of S^1, $M_3(\mathbb{C})$ at $n = 3$ from the Hurwitz fallback at S^5, and $\mathbb{H}$ at $n = 2$ *admitted* but not autonomously forced over $M_2(\mathbb{C})$ in the J-sign-table audit (Door 4c). The admitted-not-forced status of $\mathbb{H}$ is the lone admitted entry in our catalogue and is recorded as B5.

The Forced-Count Theorem [9] (Paper 32 §VIII, `thm:forced_count`) chains the standard real-spectral-triple axioms (Hermiticity, KO-dim-6 chirality, J-reality, order-one) into the bit-exact *matter-sector* moduli dimension

$$\dim_{\mathbb{R}} \mathcal{M}(D_F)\big|_{\text{matter}} \;=\; 128 \quad \text{per generation}, \tag{2}$$

via the matter-sector chain $1024 \to 512 \to 128 \to 128$; the full-axiom D_F moduli dimension is 260, not 128 (Paper 32). The eight free parameters per generation in the diagonal slice are exactly the Yukawa values (charged-lepton + quark sector). The theorem forces the *count*; it does not force the *values*.

The Higgs admission — the structural fact that the Mexican-hat field-theory structure is allowed within the inner factor's spectral action — is the conclusion of Sprint H1 (POSITIVE-THIN verdict, Paper 32 §VIII.C). It is a forcing of *structure*, paired with a non-forcing of *value* via Sprint H1's Yukawa non-selection theorem.

2.3 Spectral / period content (C)

The master Mellin engine of Paper 18 §III.7 [4], sharpened in Paper 32 §VIII via the case-exhaustion theorem [9], classifies every transcendental appearing in a GeoVac observable as a member of one of three sub-rings:

$$M_1 = \mathbb{Q}[\pi, \pi^{-1}] \quad \text{(Hopf-base measure, } k = 0\text{)}, \tag{3}$$

$$M_2 \subset \bigoplus_k \pi^{2k}\mathbb{Q} \quad \text{(Seeley–DeWitt heat kernel, } k = 2\text{)}, \tag{4}$$

$$M_3 \subset \mathcal{MT}(\mathbb{Z}[i, 1/2]) \text{ at level } \leq 4 \quad \text{(vertex-parity Hurwitz, } k = 1\text{)}. \tag{5}$$

The classification is theorem-grade. The signature prefactor $M_1 = \mathrm{Vol}(S^2)/4 = \pi$ recurs across the corpus, including as the universal van Suijlekom state-space Gromov–Hausdorff convergence rate $4/\pi$ that is rank-invariant across all compact Lie groups (Paper 38, Paper 40 [13, 14]).

2.4 Gravity (D)

The spectral action on S^3 is two-term-exact by the Bernoulli identity $\zeta_{\text{unit}}(-k) = 0$ for all $k \geq 0$ [17]; the framework outputs the exact closed forms

$$S_{\text{BH}} = \frac{A\Lambda^2}{12\pi}, \qquad G_N = \frac{3\pi}{\Lambda^2}, \qquad s_{\min} = -\Lambda/(12\sqrt{6}), \tag{6}$$

together with the Sommerfeld–Cheeger cone coefficient $-1/12$ and the replica derivative $d\Delta_K/d\alpha\big|_{\alpha=1} = +1/6$ at bit-exact precision (Paper 51 §G4-4c, G4-4f). The relations among G_N, Λ_{cc}, and S_{BH} are forced by Wald's theorem (two-term-exactness $\Rightarrow$ pure Einstein action $\Rightarrow$ action-$G \equiv$ entropy-G). The relations are forced; the cutoff function f itself (its $\varphi(0), \varphi(1), \varphi(2)$ moments) is not — see §3.

2.5 QED and the α coincidence (E)

Paper 28 establishes five theorems on S^3 QED [8]: the self-energy structural zero $\Sigma(n_{\text{ext}} = 0) = 0$, the squared-Dirac period ring $\zeta_{D^2}(s) \in \sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$ (T9), the χ_{-4} identity $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$, the $\zeta(3)$ complementarity, and the product survival rule. The Paper 33 vertex-promotion analysis recovers the full 1+6+1 partition of QED selection rules at the structural level [10].

The fine-structure-constant coincidence is decomposed into three independent spectral homes: $B = 42$ (Casimir trace), $F = \pi^2/6$ (Fock-degeneracy Dirichlet at the packing exponent), $\Delta = 1/40 = c^2(3,2)$ (Dirac mode degeneracy g_3^{Dirac}). Each is forced. Twelve mechanisms have been eliminated for the combination rule $K = \pi(B + F - \Delta)$ as a single-principle derivation; the combination rule itself remains an Observation and is recorded as a calibration entry in §3.

2.6 Chemistry / quantum computing scaling (H)

Paper 22's potential-independent angular-sparsity theorem [5] forces 1.44% ERI density at $l_{\max} = 3$ regardless of the potential. Paper 14's universal $N_{\text{Pauli}} = 11.10 \times Q$ holds across

38 molecules with the isostructural-invariance corollary ($CO = N_2 = F_2 = 1111$) [3]. The Pauli ratio balanced/composed grows as $O(B^2)$ at the structural-architectural level. Paper 23's Fock projection rigidity theorem [6] establishes that the S^3 conformal projection is unique to $-Z/r$ and does not transfer to the harmonic oscillator or Woods–Saxon, with downstream consequence that the nuclear shell-closure magic numbers $\{2, 8, 20, 40, 70, 112\}$ are forced by graph counting on the Hopf base.

2.7 Foundations of the operator-algebraic structure (within C)

The math.OA arc closes WH1 PROVEN (Paper 38) with the GH-convergence rate $4/\pi$ via the five-lemma proof (L1' / L2 / L3 / L4 / L5) [13]. Paper 40 extends the rate to all compact Lie groups (rank-invariant universality) [14]. Paper 43 records the Pythagorean orthogonality $\langle H_{\text{local}}, D^L_W\rangle_{\text{HS}} = 0$ at bit-exact precision [15], carrying the $1/\pi^2$ master-Mellin-engine M_1 signature into the Lorentzian extension.

2.8 Catalogue

A consolidated table of the 35 forced entries with mechanism, witness, and the three discriminator columns is given in the companion catalogue `docs/forced_free_seam.md`. The structural pattern across the forced inventory is: every entry sits at multi-focal depth MF = 1, the period-class assignment is either "none" (skeleton / gauge / counting facts), "meta" (the engine itself), or $M_1/M_2/M_3$ (specific period-content forced facts), and dimensional character is overwhelmingly dimensionless — the gravity sub-domain D2 (S_{BH}, G_N) being the prominent exception, which is itself only dimensionful by virtue of carrying the external cutoff scale Λ.

3 The free side

We organize the free inventory along the same sub-domain axes. The decisive distinction in this section is between two structurally different families of calibration content, an observation that emerges from the catalogue itself rather than being imposed on it.

3.1 Standard Model inner factor (F)

The most load-bearing free row is the Standard Model inner factor. Sprint H1's Yukawa non-selection theorem [9] establishes structurally that no inner fluctuation of D_F on $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ autonomously selects the Yukawa values $\{y_e, y_\mu, y_\tau, y_u, y_d, y_s, y_c, y_b, y_t\}$. The corresponding empirical confirmation is the 162-cell PSLQ sweep of Sprint Yukawa-PSLQ [23]: at coefficient ceiling $M \leq 1000$, at both the M_Z and the $\mu = 2 \times 10^{16}$ GeV unification scales, in three transforms (y_f, y_f^2, $\log y_f$), with the basis sharpened to $M_1 \cup M_2$ (pure-Tate, the only periods allowed on Connes–Chamseddine-compatible inner factors per Paper 18 §IV.6 `thm:eta_trivialization`), *zero hits* were returned. The clean negative across 162 cells is the strongest available calibration verdict at sprint scale.

The generation count $N_{\text{gen}} = 3$ is similarly free. Sprint C3 (2026-06-08, `debug/sprint_c3_n_gen_non_selec` upgraded the prior Direction 2 + Read 2 NO-GO scopings to a theorem-grade structural result: the N_{gen} non-selection theorem (Paper 32 `thm:n_gen_non_selection`) states that under the canonical Chamseddine–Connes SM representation, for every integer $N \geq 1$ the GeoVac spectral triple with inner Hilbert-space multiplicity N satisfies every axiom in the framework's current set $\mathcal{A}$, hence $\mathcal{A}$ contains no morphism selecting a specific value of N. The conditional on canonical representation is honest; the unconditional question (whether any SM-phenomenologically-consistent rep correlates N with algebra factor count under an alternative rep) remains the open long-range Read 2 sharpest-falsifier target. The Direction 2 NO-GO and Read 2 NO-GO [24, 25] supply the structural ingredients: (a) the current Paper 0 packing axiom does not reach the multiplicity-three structure as a consequence (Direction 2), and (b) the candidate Hopf-tower-to-representation shortcut fails on three independent grounds in the standard CCM representation (Read 2) — the 32 fermion DOF per generation are distributed across all three algebra factors, the CKM/PMNS cross-generation Yukawa coupling is structurally required, and the DOF count is wrong by a factor of 8 in the alternative reading. No published proposal in the spectral-triple literature (Krajewski 1998, Connes 2006, CCM 2007, Devastato–Lizzi 2013–2015, Boyle–Farnsworth 2018, Ma 2018, Bochniak–Sitarz 2025, Hessam–Khalkhali) derives $N_{\mathrm{gen}} = 3$ from a Hopf-tower argument; Ma 2018 derives it via an independent algebra-extension axiom, not from the existing Paper 0 input.

The inner-factor KO-dimensional signature is similarly free. Sprint F3 (2026-06-08, `debug/sprint_f3_ko_di` upgrades the prior Direction 2 packing-reach NO-GO scoping into a theorem-grade structural result: the inner KO-dim non-selection theorem (Paper 32 `thm:ko_dim_non_selection`) composes Paper 0 §VII.B's explicit kinematic-scope statement (packing produces labels and graph topology, but no Dirac operator, no real structure, no chirality grading) with the Door 4f / Door 4b finding that inner KO-dim is a property of (D_F, J_F, γ_F). The composition is one line: $\mathcal{A}$ produces no (D_F, J_F, γ_F) data autonomously, hence cannot autonomously select KO-dim. The conditional is again the canonical CCM rep; the unconditional version (whether any non-canonical SM-phenomenologically-consistent rep yields a different KO-dim) remains open. The CKM and PMNS mixing-angle entries inherit F1 + F4 structurally (Boyle–Farnsworth gives the shape but not the values). The Higgs VEV v and the neutrino mass values are dimensionful scales consumed externally.

3.2 Multi-focal-composition walls (G; cf. §4)

Six independent observables converge on the same structural reason for being free. We catalogue them here and discuss their unity in §4:

1. **HF-3 recoil cross-register** $V_{eN}(\hat{r}_e, \hat{R}_n)$: electron–nucleus potential as a two-body spatial coupling. The framework couples discrete labels (electron spin, nuclear spin) cleanly via Wigner symbols, but the spatial composition of two Fock-style projections (electron register, nucleus register) is structurally outside the framework's native composition apparatus.

2. **HF-4 Zemach magnetization density**: proton magnetization × electron $1s$ wavefunction is a spatial × spatial composition. The W1b extension is operator-level partial; the magnetization-density profile itself is external.

3. **HF-5 multi-loop QED** on hyperfine: bare two-loop vertex topologies diverge $\sim N^{3.79}$ with no autonomous renormalization, structurally the same wall as LS-8a.

4. **LS-8a two-loop SE** renormalization: iterated Chamseddine–Connes spectral action reproduces the two-loop UV-divergent integrand faithfully (right prefactor, sign, divergence) but cannot autonomously generate the Z_2 and δm counterterms.

5. **W1e chemistry** inner-region overattraction: the chemistry-side analog of the Yukawa non-selection theorem [26]. Six framework-internal modification attempts (F1–F6 of CLAUDE.md §3, plus Schmidt, core correlation, basis enlargement, explicit-core HF, DMRG) have failed; the W1e period-class diagnostic returns zero genuine outer-factor (M1/M2/M3) identifications across 11 correction terms.

6. **H1 Yukawa non-selection** (= F1): the SM-side instance.

All six entries sit at multi-focal depth MF > 1, period class "outside" the master Mellin engine, and dimensional character mostly dimensionful (the exception being the Yukawa values themselves, which are dimensionless but lie outside the inner-factor period ring).

3.3 Gravity calibration (D5–D6)

The Mellin moments $\varphi(0)$, $\varphi(1)$, $\varphi(2)$ of the cutoff function f are externally supplied. The sector-wise moment map (tip $\leftrightarrow \varphi(0)$, Einstein–Hilbert $\leftrightarrow \varphi(1)$, $\Lambda_{cc} \leftrightarrow \varphi(2)$) is proven and the relations among the gravitational observables are forced (D7), but the moment *values* are not. The cosmological-constant problem in GeoVac is structurally precise: obtaining the observed dimensionless ratio $\Lambda_{cc} \cdot G_{\text{eff}} \approx 36\pi \cdot \varphi(2)/\varphi(1)^2 \sim 10^{-124}$ requires a cutoff function with $\varphi(2)/\varphi(1)^2 \approx 10^{-124}$ while $\varphi(0), \varphi(1) = O(1)$, which no natural cutoff produces. This is a calibration selection problem on the cutoff function, structurally similar to the Higgs–Yukawa selection problem but on a different object.

3.4 Chemistry-side inner-factor input data (H6–H7)

The Clementi–Raimondi 1963 atomic-Hartree–Fock fits supply the screening exponents $\zeta_{1s}, \zeta_{2s}, \zeta_{2p}$ used in the FrozenCore $Z_{\text{eff}}(r)$ profile for second-row and heavier chemistry. The multi-zeta valence basis coefficients are similarly external. The W1e period-class diagnostic [26] classifies these as the chemistry-side analog of the SM inner-factor input data (Paper 18 §IV.6 sixth-tier reading): the framework provides the multi-electron Hamiltonian assembly machinery (composed_qubit, balanced_coupled) on top of these inputs but does not autonomously select the inputs. The chemistry analog of η-trivialization (Paper 18 §IV.6, `thm:eta_trivialization`) is the open structural question on this side.

3.5 Foundational calibration (Class 1; I)

Three entries sit at the foundational level. The fine-structure constant α inherits E6 (the combination rule is an Observation; the value is external). The Born rule probability assignment $p = |\langle a|\psi\rangle|^2$ is structurally external: the framework uses Gleason's theorem via Hilbert-space inheritance and does not improve on Gleason. The Higgs direction $\hat{n} \in S^2$ in the Boyle–Farnsworth parameterization [19] is similarly external, with an open structural question of whether $\hat{n}$ should be identified with the GeoVac Hopf-base S^2 (carrying the M_1 signature $\text{Vol}(S^2)/4 = \pi$) — a flagged-but-not-pursued follow-on.

3.6 Two structurally distinct families on the free side

The catalogue reveals an observation that was anticipated but never sharply tested in the prior literature.

Observation 3.1 (Two-family structure of calibration content)**.** The free-side entries decompose into two structurally distinct families with different discriminator-column signatures:

1. **Multi-focal composition family** (G1–G6, E7, E8, F1, F4, F7, D5, D6): "the framework can build it but cannot autonomously pick a number." Signature: MF > 1, period class outside, mostly dimensionful.

2. **Inner-factor input data family** (F2, F3, B8, H6, H7, I2, I3): "categorical to the framework's outer-factor mechanism; lives elsewhere by structure." Signature: MF $= 1$, period class "none" or N/A, dimensionless.

This is the operative refinement of the structural-skeleton scope pattern. The earlier one-bucket framing ("structural skeleton vs. calibration data") was correct but coarse. The two-family observation aligns with the previously-named external-input three-class partition, with the third class (multi-determinant correlation) appearing only in the chemistry sub-domain as W1e and currently best read as a sub-case of family 1 with chemistry-specific composition machinery.

4 Cross-domain unity: the multi-focal-composition wall

The most structurally striking pattern in the catalogue is that six observables across four sub-domains converge on a single structural reason for being free.

4.1 The six instances

Listed in order of historical surfacing:

1. **H1 (Sprint H1, 2026-05-31; SM-side)**. $\mathcal{A}_F$ admits the Higgs structurally but does not autonomously select Yukawa values.
2. **LS-8a (2026-05; QED two-loop SE)**. Iterated Chamseddine–Connes spectral action reproduces the two-loop integrand structure but cannot generate $Z_2, \delta m$ counterterms.
3. **HF-3 (Sprint HF, 2026-05-07)**. Track NI cross-register couples discrete spin labels but does not couple spatial coordinates between electron and nucleus registers.
4. **HF-4 (Sprint HF, 2026-05-07)**. hyperfine_coupling_pauli is point-like in proton position; no magnetization-density distribution carried.
5. **HF-5 (Sprint HF, 2026-05-07)**. Bare two-loop vertex topologies diverge $\sim N^{3.79}$ with no autonomous renormalization (cf. LS-8a).
6. **W1e (Sprint W1c arc, 2026-05-23)**. NaH inner-region overattraction; six framework-internal modification attempts fail at sprint scale.

4.2 The unifying statement

Observation 4.1 (Multi-focal-composition wall)**.** The framework couples discrete labels (quantum numbers $n, l, m, m_s, \kappa, m_j$, isospin, color, gauge representations) cleanly via Wigner symbols and selection rules. Single-focal-length QED on S^3 is handled including curvature corrections at structural level (Parker–Toms verified at 0.5% in HF-2). Spatial coordinates exist only after a Fock-style projection fixes a focal length; multi-focal observables that require composing two or more such projections at once (electron–nucleus V_{ne} with quantum-mechanical nucleus position, proton magnetization $\times$ electron wavefunction, electron–electron r_{12} where each electron carries its own scale, multi-loop QED renormalization between bare integrand and counterterm structure) lie outside the framework's native composition apparatus.

The G4b cross-manifold question $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}$ is the spectral-triple-level instance of the same wall. Paper 24 §V records this as the fourth Coulomb / HO asymmetry layer [7].

4.3 Operational consequence

When dispatching new investigations, the catalogue affords a forward-classification:

- **Discrete-label couplings** $\rightarrow$ framework handles natively. Single sprint, expect positive.
- **Single-focal-length QED on S^3** $\rightarrow$ framework handles natively; curvature corrections expected (Parker–Toms 0.5%).
- **Multi-focal spatial composition** $\rightarrow$ framework hits Observation 4.1. Sprint should be diagnostic, naming which composition is missing, not implementation. Expect negative-with-clear-naming.
- **Renormalization counterterms** $\rightarrow$ framework hits the LS-8a wall (= Observation 4.1 sub-case).

This classification is what permits the catalogue to function as a research-direction tool rather than a static record: candidate investigations can be pre-classified as Class A (handle), Class B (handle, light corrections), Class C (diagnostic only), Class D (out of scope at present axiom level) before resources are committed.

5 Candidate discriminator principles: a preliminary probe

This section probes four candidate principles for what distinguishes forced from free content. None of them is established as *the* principle. They are tested against the catalogue and reported on; the analysis is preliminary by design.

5.1 Candidate P1: multi-focal depth

Principle. Forced iff MF $= 1$; free iff MF > 1.

Coverage. Every multi-focal-composition wall entry (G1–G6, plus E7, E8, F1, F4, F7, D5, D6) sits at MF > 1, correctly placed as free. Every forced entry except D2 sits at MF $= 1$.

Misclassifications. Five entries break P1 in the forced direction: F2 (N_{gen}), F3 (KO-dim), B8 (gauge lower bound), H6–H7 (chemistry calibration), I2 (Born rule), I3 (Higgs direction) — all MF $= 1$ but free. These are precisely the entries of the *inner-factor input data* family (Observation 3.1, item 2). P1 catches the multi-focal composition family cleanly but is blind to the inner-factor input data family.

Verdict. P1 is a one-family discriminator (it discriminates the multi-focal composition family) but does not generalize. The empirical evidence is consistent with the two-family structure of Observation 3.1: multi-focal depth is the right axis *for* the multi-focal composition family, full stop.

5.2 Candidate P2: period-content classifiability

Principle. Forced iff the entry is classifiable in the master Mellin engine $M_1/M_2/M_3$ ring; free iff it sits outside.

Coverage. The multi-focal walls all sit at "outside," correctly placed as free. The forced spectral / period entries (C1–C9) sit at $M_1/M_2/M_3$ or "meta," correctly placed as forced.

Misclassifications. P2 is structurally wrong for the gauge / counting forced entries (A1–A5, B1–B7), most of which sit at period class "none." They are forced not because they live in the period ring but because they are non-period combinatorial / representation-theoretic facts. P2 also misclassifies F2, F3 in the same way: "period class none" applies to both forced counting facts and to free inner-factor input data.

Verdict. P2 catches the multi-focal composition family but is the wrong axis for the rest of the forced inventory and for the inner-factor input data family.

5.3 Candidate P3: dimensional character

Principle. Forced iff dimensionless; free iff dimensionful.

Coverage. Most forced entries (skeleton, gauge, periods, chemistry/QC scaling) are dimensionless ratios. Most multi-focal-wall free entries are dimensionful scales (cutoff f, Λ_{cc}, v, etc.).

Misclassifications. D2 (S_{BH}, G_N) is forced and dimensionful. F1 (Yukawa values), F2 (N_{gen}), F3 (KO-dim), F5 (CKM), F6 (PMNS), B8 (gauge lower bound) are all free and dimensionless. Six counterexamples on each side.

Verdict. P3 catches a correlation but not a principle. The correlation reflects that dimensionful objects often require a scale-supplying calibration step, but it does not capture the structural source of the free / forced split.

5.4 Candidate P4: compactness inheritance

Principle. Forced iff derivable from the compact / Peter–Weyl regime of the Paper 0 packing axiom; free iff requires continuum (un-compact) data.

Coverage. The skeleton / spectral / Mellin-engine forced entries (A, C) derive from Peter–Weyl on compact S^3, correctly placed as forced. The cutoff function f, the Higgs VEV v, the multi-loop counterterms are continuum-side data, correctly placed as free.

Misclassifications. Same blind spot as P1 / P2 in a slightly different language: the inner-factor input data family (F2, F3, H6, H7, I2) is not obviously continuum data, yet it is free. P4 also has to be made precise — the boundary between "derivable from Peter–Weyl on compact S^3" and "derivable from compact data, broadly" is not currently sharp enough to test cell-by-cell.

Verdict. P4 captures the structural-skeleton scope reading of CLAUDE.md §1.7 WH3 cleanly for the multi-focal composition family (continuum data = un-compact = free), but fails on the inner-factor input data family and needs sharpening to become a formal cell-by-cell predicate.

5.5 Candidate P5: packing-reachability

The four candidates above all fail on the inner-factor input data family. Sprint C2 (2026-06-08, `debug/sprint_c2_principle_hunt_memo.md`) introduces a fifth candidate that does not.

Principle. An entry is *packing-reachable* iff there exists a witness structural derivation from the axiom set

$$\Big\{\text{Paper 0 packing axiom; standard real-spectral-triple axioms (Hermiticity, } J\text{-reality, KO-dim chirality, order-o}$$

$$\text{Hopf-tower truncation (Hurwitz); Bertrand's classical theorem; Upgrade B (sphere-Lie-group axiom adopted);}$$

$$\text{CCM canonical SM representation on } \mathcal{H}_F\Big\}.$$

Predict forced iff packing-reachable.

Coverage. Every forced entry in the catalogue carries a witness derivation in the corpus: Paper 32 §VIII for the gauge group and the Forced-Count Theorem [9]; Paper 18 for the master Mellin engine [4]; Paper 28 for the QED structural theorems [8]; Paper 38 for GH convergence [13]; Paper 51 for the gravity closed forms [17]; Papers 22, 14, 23 for the chemistry / QC / nuclear scaling structurals [3, 5, 6]. Every calibration entry carries either an explicit non-selection theorem (Sprint H1's Yukawa non-selection; the W1e period-class diagnostic [26]) or a documented no-go / negative-search memo (Direction 2 and Read 2 NO-GOs on N_{gen} and inner KO-dim [24, 25]; LS-8a renormalization wall [12]; 162-cell Yukawa-PSLQ clean negative [23]).

Misclassifications. One entry, I3 (Higgs direction $\hat{n} \in S^2$), is tagged *conditional*: the Boyle–Farnsworth Higgs unit vector could in principle be identified with the GeoVac Hopf-base S^2 (carrying the master Mellin engine M_1 signature $\mathrm{Vol}(S^2)/4 = \pi$), an open follow-on flagged but not pursued. Treating conditional as forced gives one false positive (I3 predicted forced, actual calibration). The misclassification is precisely the open identification, not a defect in P5.

Verdict. P5 is the *consolidated statement of the forced/free boundary itself, not an independent discriminator of it.* An entry is tagged packing-reachable exactly when the corpus carries a forcing derivation, and packing-unreachable exactly when it carries a non-selection theorem or documented negative search — but "forcing derivation exists" is the *definition* of Forced and "non-selection theorem / negative search" is the definition of Calibration, so the packing-reachable tag is, by construction, the F/C label under another name. The 98.3% agreement with the F/C status (59/60; the lone exception I3 is deliberately tagged *conditional* to encode an open identification) is therefore an *internal-consistency check* — it verifies the catalogue applied its own F/C conventions uniformly across 60 entries — not a discovered structural predictor, and we do not present it as one. The genuine discriminator content is carried by multi-focal depth (P1), period class (P2), and the two-family decomposition (Observation 3.1), all tagged independently of the F/C status; P5 organizes that content rather than predicting it (see §6.1).

What P5 unifies. P5 absorbs P1, P2, and P4 as special cases. Family-1 entries fail P5 because the framework lacks a multi-focal composition theorem (MF $> 1 \to$ no closure $\to$ packing-unreachable). Family-2 entries fail P5 because their values live in a categorical-external register on which the outer-factor mechanism is structurally silent (η-trivialization on the SM side, Paper 18 §IV.6; chemistry-side analog at the inner-factor input-data tier). The two families exhibit different reasons for failing P5, but both fail it.

5.6 Synthesis after five candidates

Table 1: Candidate principles after Sprint C2. Accuracy is true-positive + true-negative over total entries on the 60-entry catalogue. "Family 1" = multi-focal composition (17 calibration entries); "Family 2" = inner-factor input data (7 calibration entries). P5 is the consolidated F/C boundary (definitionally the status column); its 98.3% is an internal-consistency check, not independent accuracy (§6.1).

Principle	Accuracy	Family 1	Family 2	Misclass.
P1 multi-focal depth	86.7%	✓	misses	8
P2 period classifiable	86.7%	✓	misses	8
P3 dimensional character	76.7%	partial	misses	14
P4 compactness inheritance	96.7%	✓	one (open)	2
P5 packing-reachability	**98.3%**	✓	✓	**1 (open)**
P1 $\wedge$ P5	96.7%	✓	one (open)	2
P1 $\vee$ P5	88.3%	✓	misses	7

P5 (packing-reachability) is the *consolidated restatement* of the forced/free boundary, not an independent fifth discriminator: its 98.3% agreement with the F/C status is an internal-consistency check (the packing-reachable tag is the F/C label by construction; §6.1). The genuine discriminator content is the two-family decomposition and the independently-tagged multi-focal-depth and period-class axes.

Observation 3.1's two-family decomposition is preserved as the structural failure-mode story beneath P5, not as evidence that two separate principles are needed. The two families share the surface verdict (packing-unreachable) but exhibit different structural reasons for it, with η-trivialization on the inner-factor side (family 2) and multi-focal-composition gap on the projection-apparatus side (family 1).

The honest interpretation: the forced/free seam IS the packing-reachable / packing-unreachable boundary, and P5 *names* that boundary rather than independently predicting it. The sharpened open question — whether the boundary admits a *meta-theorem* that decides forcedness without first locating the witness — is §6.

6 Open questions

6.1 P5 and the tautology question (resolved)

Sprint C2 (§5.5) introduced packing-reachability and left open whether it is a structural principle or is definitionally equivalent to the F/C label.

The tautology concern. The packing-reachable tag for each entry was determined by examining the corpus for witness derivations (forced entries) and explicit non-selection theorems /

negative-search memos (calibration entries). If "packing-reachable" is operationally just "we have a derivation," then P5 = F/C label by construction, and the 98.3% agreement is an internal-consistency check rather than an independent verdict.

A direct check settles it. We attempted to determine packing-reachability by an *F/C-blind* criterion — "does the corpus carry a cited theorem or bit-exact test for this entry?" — applied identically to every entry. It does not discriminate: all 60 entries carry such a witness (the forced entries via forcing derivations, the calibration entries via the non-selection theorems of Paper 32 §VIII and the negative-search memos), so an F/C-blind witness-existence check predicts "forced" for all 60 and scores at the 60% base rate. The 98.3% is recovered only by reading each witness's *polarity* (forcing vs. non-selection), and that polarity reading *is* the F/C judgment. We therefore conclude that P5 **cannot be made non-circular**: it is the forced/free boundary restated, and its agreement with the F/C label is an internal-consistency check. This does not weaken the catalogue — the structural discovery is the two-family decomposition and its independent multi-focal-depth / period-class signatures (§5), which P5 consolidates.

The sharpened open question. The open question is correspondingly sharpened: not "is P5 tautological" (it is), but whether the packing-reachable boundary admits a *meta-theorem* — a structural characterization deciding forcedness *without* first locating the forcing-or-non-selection witness. A theorem of the form "any observable on the GeoVac spectral triple is forced iff packing-reachable" would lift the boundary from internal bookkeeping to structural truth. It is long-range, NCG-research-grade, and named here as the sharpest follow-on of the Sprint C2 finding. A first answer — covering the skeleton, the inner-factor family, the admitted value, and, at mechanism level on both corpus-faithful instances, the multi-focal family — is the Bohr-site dichotomy of §6.2.

6.2 A Bohr-site partial meta-theorem (Sprint Topos-1)

For a finite-dimensional C*-algebra A, let $\mathcal{C}(A)$ denote the poset of commutative unital *-subalgebras (the *Bohr site* of the topos approach to quantum logic). Sprint Topos-1 tested the candidate meta-theorem: an entry is **forced** iff it is a property of the site (plus the distinguished GeoVac flag) requiring no valuation data; **calibration** iff it is valuation data — a point the site cannot supply; **admitted** iff the site is *degenerate* (cannot distinguish the candidates); and the *dimension* of each valuation freedom is itself a site-side fact. Four computed legs (backing: `tests/test_topos1_site_invariants.py`, bit-exact):

1. **Site strata** (INTERNAL THEOREM, combinatorial): conjugacy strata of commutative unital *-subalgebras of $M_k(\mathbb{C})$ correspond to integer partitions of k with stratum family dimension $k^2 - \sum_i \lambda_i^2$: M_2: $\{0, 2\}$; M_3: $\{0, 4, 6\}$; $\mathbb{H}$ (real form): $\{0, 2\}$ (the $\mathbb{C}_q$ family over $S^2/\{\pm\}$).
2. **The B5 correspondence** (MEASURED, order-invariant level): $\mathcal{C}(M_2(\mathbb{C}))$ is the height-1 flat fan (one point stratum plus one 2-parameter family of maximal 2-dimensional subalgebras, moduli $\mathbb{R}P^2$, with no intermediate elements) — the minimal non-rigid Boolean block, which is *precisely* the excluded case of the Döring–Harding / Hamhalter reconstruction theorems [20, 21]: an order isomorphism $\mathcal{C}(A) \to \mathcal{C}(B)$ is induced by a Jordan *-isomorphism for all A with *no type I_2 summand* (and $A \neq \mathbb{C} \oplus \mathbb{C}$), and type I_2 is exactly $M_2(\mathbb{C})$. At $k = 3$ the appearance of intermediate subalgebras (height 2) restores this rigidity. We observe that $\mathcal{C}(\mathbb{H})$, for the real *-algebra $\mathbb{H}$ whose complexification is $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$, shares this height-1 fan and the same $\mathbb{R}P^2$ maximal-element moduli, so the two candidates at the catalogue's lone admitted-not-forced entry (B5, $\mathbb{H}$ vs $M_2(\mathbb{C})$, Door 4c) are indistinguishable by these order invariants; we record this $M_2(\mathbb{C})/\mathbb{H}$ coincidence as a framework observation (not located in the Bohrification literature, which treats

complex C*-algebras). We do *not* claim the poset fails to determine $M_2(\mathbb{C})$ as a complex C*-algebra — for finite-dimensional complex A, $\mathcal{C}(A)$ determines A up to *-isomorphism with no exception [22] — only that the Jordan-induction / automorphism-rigidity reconstruction degenerates at type I_2, which is exactly the $k = 2$ slot where the catalogue admits $\mathbb{H}$ over $M_2(\mathbb{C})$. This is decided from site properties without consulting any witness derivation.

3. **Machine-verified Kochen–Specker witness** (bit-exact): the orthogonality hypergraph of primitive integer rays in $\mathbb{R}^3$ with entries in $\{-2, \ldots, 2\}$ (49 rays, 26 orthonormal triples; containing the Conway–Kochen ray family) admits *no* 0/1 frame-function coloring (exhaustive search), while the range-1 set (13 rays) is colorable. Hence dim-≥ 3 blocks are valuation-*obstructed* (their values are necessarily external data), dim-2 blocks are valuation-*choiceable*, dim-1 blocks are pointed — three site statuses matching forced / admitted / calibration structure at the block level.

4. **The GeoVac flag** (bit-exact, $n_{\max} = 2, 3$): the Casimir chain $\langle n\rangle \subset \langle n, L^2\rangle \subset \langle n, L^2, L_z\rangle$ is a maximal chain of commutative subalgebras of dimensions $(2, 3, 5)$ and $(3, 6, 14)$; the graph adjacency satisfies $[A, L^2] = 0$ exactly (the edge rule conserves l — a selection rule that is itself a site fact) with $\|[A, n]\|, \|[A, L_z]\| > 0$.

5. **Family-1 mechanism: frame-meet collapse** (bit-exact, Sprint Topos-2): for two same-center hydrogenic Fock frames at exponents $Z = 1$ and $Z' \in \{2, 3\}$ (the mixed-exponent setting of the Track-DF / Löwdin wall), the exact rational inter-frame overlap is (l, m)-block-diagonal with *connected* support in every block ($n_{\max} \leq 4$), so the meet of the two frame MASAs is *exactly the angular-grading algebra* (collapse $14 \to 9$ and $30 \to 16$): frame composition preserves the angular structure and nothing else — the site statement of the angular-sparsity / radial-wall seam (Papers 22, 32). The join generates the full matrix algebra on each (l, m) block (connected support + irreducibility), and blocks of dimension ≥ 3 appear at $n_{\max} \geq 3$, so composed radial observables land in Kochen–Specker-obstructed algebras: **Family-1 externality = meet-collapse followed by join-obstruction** — a provably different *route* to valuation-need than Family-2's ab-initio noncommutativity, with the same terminus (matching the two-family decomposition: same surface verdict, different structural reasons). By-product exact lemma: rate coincidence $Z_1/n_1 = Z_2/n_2$ with $|n_1 - n_2| \geq 2$ forces a vanishing inter-frame overlap (same-variable Laguerre band structure; verified exhaustively for $Z' \leq 5$, $n \leq 6$), while the converse is panel-scoped (one sporadic off-coincidence zero at $Z' = 2$, $l = 3$, $n = n' = 5$). Backing: `tests/test_topos2_family1_meetjoin.py`.

6. **Two-center meet: the m-grading exactly** (exact, Sprint Topos-3): for frames at centers separated by R along z ($Z' \in \{1, 3\}$, $R \in \{1, 2\}$, $n_{\max} = 3$), every inter-frame overlap reduces in prolate spheroidal coordinates to $e^{-p}(Ue^{q} + Ve^{-q})$ with U, V exact rationals (Mulliken–Ruedenberg auxiliary integrals; zero-decidability by the linear independence of $e^{\pm q}$ over $\mathbb{Q}$; machinery certified against the classical $\langle 1s|1s\rangle(R) = (1 + R + R^2/3)\, e^{-R}$ as an exact rational identity at $R = 1, 2, 7/2$). Every m-block has *full* exact support — the same-center rate-coincidence zeros do not survive displacement — so the two-center meet is *exactly the m-grading algebra*, confirming the prediction pre-registered above. The resulting **meet-dimension ladder** $14 \to 9 \to 5$ (identical frames $\to$ same-center different focal $\to$ two-center, $n_{\max} = 3$) measures the symmetry of the composition, and the two-center join is *more* Kochen–Specker-obstructed (blocks of dimension 6 and 3) than the same-center case (dimension 3 only) — the site restatement of cross-center composition being the framework's hardest wall. Backing: `tests/test_topos3_exact_meet.py`.

7. **General-parameter vanishing lemma** (exact, Sprint Topos-4): the by-product lemma of item 5 is closed in general form. For same-l inter-frame overlaps the exact unnormalized overlap factors as $S =$ (positive prefactor) $\cdot\, P_{n_1,n_2,l}(t)$ in the rate variable $t = (Z_1/n_1 -$

$Z_2/n_2)/(Z_1/n_1 + Z_2/n_2) \in (-1,1)$, with P an explicit polynomial of degree $D = n_1 + n_2 - 2l - 2$ (for $n_1 = n_2$, P is even in t by frame-swap symmetry; for the single-radial-node family $n_1 = n_2 = l+2$, where $D = 2$, it is exactly the symmetric Jacobi / Gegenbauer polynomial $P_2^{(l,l)}$, but for higher D it is not in general a classical Jacobi), so $S = 0 \Leftrightarrow P(t) = 0$. The roots fall into three structural families: (i) $t = 0$ (rate coincidence), a root *exactly* when $|n_1 - n_2| \geq 2$ (proved over the n, l grid — the structural form of item 5's exhaustively-verified ($\Leftarrow$) direction); (ii) $t = (n_2 - n_1)/(n_2 + n_1)$ (equal-exponent hydrogenic orthogonality $\langle n_1 l | n_2 l \rangle = 0$), always a root for $n_1 \neq n_2$; and (iii) residual roots, generically *irrational*. The "sporadic" off-coincidence zeros are exactly the *rational* residual roots: in the scan $Z, n < 7$ there are precisely two families, $(n_1, n_2, l) = (5,5,3)$ with $P = 1 - 9t^2$ (roots $\pm 1/3$, the item 5 boundary case, realised at $(Z, Z') = (1,2)$) and $(4,5,2)$ with roots $\pm 1/2$; the integer zero census fully accounts $253 = 37 + 210 + 6$ over the three families. Individual-element zeros do not disturb the block *connectivity* on which items 5–6 rest. Backing: `tests/test_topos4_vanishing_lemma.py`.

Under this dichotomy the four theorem-grade non-selection results of §6.3 re-read as site statements (the moduli *dimension* 128/260 is a site-side fact; the moduli *point*, the multiplicity factor, and the real-structure signature are valuations), and the lone P5 exception I3 factorises: the Higgs direction splits into an internal *space* (the Hopf-base S^2, an M_1 site object) and an external *point* (the valuation $\hat{n}$) — the "conditional" tag is precisely the open question whether the internal space is the GeoVac S^2.

The Family-1 census (Sprint Topos-4). Mapping all 17 multi-focal (MF > 1) calibration entries onto the site mechanism shows that *Family 1 is not mechanism-homogeneous.* The meet-collapse $\to$ join-obstruction mechanism (items 5–6) is the Bohr-site image of the *spatial-composition sub-sector specifically* (G1, G2, G5; Paper 32 Theorem `thm:spatial_composition_radial_wall`) — only 3 of the 17 entries. The other 14 reach externality by *valuation, not composition*: six via a test-function / cutoff valuation (E7, E8, G3, G4 renormalization; D5, D6 cutoff moments; Theorems `thm:single_cutoff_spectral_action`, `thm:cutoff_function_external`), and eight via a point / scale / period valuation (F1 Yukawa and its duplicate catalogue label G6; F4, F5, F6, F7; E6, I1). The MF > 1 tag that defines Family 1 is therefore orthogonal to the site-mechanism split: at the site level the two valuation routes carry the *same* status as Family 2 (valuation the site cannot supply), so the operative site taxonomy is **meet-collapse** (spatial-frame composition) **vs. valuation** (everything else, Family 2 included). This sharpens item 5's headline from "Family-1 externality = meet-collapse" (an over-generalisation from the two computed spatial instances) to its precise form: spatial-composition-sub-sector externality = meet-collapse.

Honest scope. The dichotomy is a *partial* meta-theorem: it covers the skeleton (commutative $\mathcal{A}_{\mathrm{GV}}$, Bohr topos Boolean), the inner-factor family (Family 2, via the Kochen–Specker obstruction), the admitted value (site degeneracy), the valuation-freedom dimensions, and — at mechanism level on the three spatial-composition entries — the meet-collapse route of the multi-focal family, all decided from site properties without locating witnesses (the non-circularity P5 could not achieve). The general-parameter vanishing lemma (item 7) and the 17-entry census (above) are now closed; the relation to general site-reconstruction theorems at the exceptional dimension $k = 2$ is stated affirmatively in item 2 above (Döring–Harding / Hamhalter type-I_2 exception [20, 21], with the non-degeneracy of the complex isomorphism class kept explicit [22]). The full 60-entry classification (upgrading the F/C column to a site-derived column) and internal-logic statements of the eight non-selection theorems remain open catalogue follow-ons. Nothing here touches the Observation status of the combination rule K: its *value* classifies as an external valuation (route 3 of the census), consistent with and not upgrading that label.

6.3 Sprints C3 + F3 closure: four theorem-grade non-selection results

Sprints C3 + F3 (2026-06-08, this same close-of-day) upgraded the F2 (N_{gen}) and F3 (inner KO-dim) entries from diagnostic-grade to theorem-grade via Paper 32 `thm:n_gen_non_selection` and `thm:ko_dim_non_selection`. The corpus now carries four theorem-grade non-selection results characterising the free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level:

1. **Forced-Count Theorem** (Paper 32 `thm:forced_count`): the matter-sector moduli dimension $\dim \mathcal{M}(D_F)|_{\text{matter}} = 128$ per generation is forced (the full-axiom D_F moduli dimension is 260; Paper 32).

2. **H1 Yukawa non-selection theorem** (Paper 32 §VIII.C): no inner fluctuation of D_F autonomously selects the Yukawa values within $\mathcal{M}(D_F)$; the diagonal slice carries 8 free parameters per generation.

3. N_{gen} **non-selection theorem** (Paper 32 `thm:n_gen_non_selection`, Sprint C3): no axiom in the framework's current set selects a specific N_{gen}; the multiplicity factor is independent of the axiom set under canonical representation.

4. **Inner KO-dim non-selection theorem** (Paper 32 `thm:ko_dim_non_selection`, Sprint F3): the framework's axiom set is kinematic (Paper 0 §VII.B) and KO-dim is real-structure data; $\mathcal{A}$ cannot autonomously select an inner KO-dim signature.

Together, these four theorems characterise the full free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level (Paper 32 `rem:full_inner_factor_boundary` the forced data is the moduli space and its dimension (Forced-Count); the free data is (i) the point in the moduli space (Yukawa values, H1 non-selection), (ii) the multiplicity factor (N_{gen}, C3 non-selection), and (iii) the real-structure signature (inner KO-dim, F3 non-selection). All three free data are calibration inputs in the structural-skeleton-scope sense.

The remaining empirical-only non-selection entries (E6 combination rule K; G1–G2 spatial composition walls; G5 chemistry composition; D5–D6 CC fine-tuning; F4–F7 Higgs VEV / CKM / PMNS / neutrino masses; H6–H7 chemistry inner-factor data; I1–I3 foundational calibration) remain candidates for future theorem-grade upgrades, but they sit outside the inner-factor sector that the four theorems above now close.

Sprint E7/E8 closure: multi-loop QED renormalization sub-sector. A fifth theorem-grade non-selection result was added in Sprint E7/E8 (also 2026-06-08), this time in the spectral-action sector rather than the inner-factor sector. Paper 32 `thm:single_cutoff_spectral_action` states that the Chamseddine–Connes spectral action $S(D, \Lambda) = \operatorname{Tr} f(D/\Lambda)$ in the framework's axiom set $\mathcal{A}$ is single-cutoff; no morphism derivable from $\mathcal{A}$ produces a multi-cutoff structure. The corresponding corollary (`cor:multi_loop_renormalization_wall`) classifies catalogue entries E7 (two-loop SE counterterms $Z_2, \delta m$), E8 (multi-loop QED beyond LS-7), G3 (HF-5 multi-loop hyperfine), and G4 (LS-8a, = E7) as multi-cutoff renormalization calibration data. The theorem formalises the LS-8a wall observed empirically in Paper 36 LS-8a (May 2026): the framework reproduces the bare two-loop integrand but cannot autonomously generate the renormalization counterterms.

Sprint G1/G2/G5 closure: spatial-composition sub-sector. A sixth theorem-grade non-selection result was added in Sprint G1/G2/G5 (same day, 2026-06-08), closing the remaining spatial-composition sub-sector of the multi-focal-composition wall. Paper 32 `thm:spatial_composition_rac` states that in the framework's tensor-product spectral triple $\mathcal{T}_a \otimes \mathcal{T}_b$, the angular structure of composed two-body operators is forced (Paper 54 Thm 3: Gaunt selection rules, m-conservation, monopole hierarchy autonomously reproduced) but the radial coupling profile is not autonomously matched to specific physical two-body forms. The structural reason is the non-commutativity of the Fock-projection conformal factor (Paper 7 chordal-distance identity) with the tensor-product construction: two focal lengths give two different conformal factors c_a, c_b whose combined product $c_{ab} \neq c_a \cdot c_b$, and the relative-coordinate reduction from the 6-dim product manifold to the 3-dim relative manifold is not in $\mathcal{A}$.

The corresponding corollary (`cor:spatial_composition_wall`) classifies catalogue entries G1 (HF-3 recoil cross-register V_{eN}), G2 (HF-4 Zemach magnetization), G5 (W1e chemistry inner-region) as spatial-composition calibration data: angular content forced, radial content external. Empirical verification comes from Sprint resolvent_two_body (2026-06-01) which tested four resolvent weightings against Coulomb and found Pearson correlations ≤ 0.81 decreasing with $n_{\max}$.

The multi-focal-composition wall is now fully theorem-bound. Paper 32 `rem:multi_focal_wall_full` records the closure of all six unique multi-focal-composition wall instances (G6 = F1 is not independent) at theorem-grade level under two structurally distinct theorems:

- **Renormalization sub-sector** (E7, E8, G3, G4): closed by Theorem `thm:single_cutoff_spectral_act`
- **Spatial-composition sub-sector** (G1, G2, G5): closed by Theorem `thm:spatial_composition_radial`

Six theorem-grade non-selection results in the corpus. After a single sprint wave (Sprints C3 + F3 + E7/E8 + G1/G2/G5 closures, all 2026-06-08):

1. Forced-Count Theorem (inner factor moduli dimension)
2. H1 Yukawa non-selection (inner factor diagonal slice point)
3. N_{gen} non-selection (inner factor multiplicity)
4. Inner KO-dim non-selection (inner factor real-structure signature)
5. Single-cutoff spectral action (multi-loop renormalization sub-sector)
6. Spatial-composition radial wall (spatial-composition sub-sector)

The first four close the inner-factor structural-skeleton boundary at the canonical-rep level. The next two close the multi-focal-composition wall family.

Sprint C-arc closure (E6 + D5/D6 + chemistry analog). Two additional theorem-grade non-selection results and one named substantial structural-research target were added in the C-arc terminal-state sprint (same day, 2026-06-08, v3.103.0), bringing the corpus to *eight* theorem-grade non-selection results plus one named substantial target:

7. **No-single-mechanism** $K = \pi(B + F - \Delta)$ (Paper 32 `thm:no_single_mechanism_K`, Sprint E6): no morphism in $\mathcal{A}$ produces the empirical combination rule for α^{-1} as a single-mechanism output; three spectral homes live in two distinct Mellin sub-rings; twelve mechanisms empirically eliminated (Phases 4B–4I + Sprint A α-SP + Sprint K-CC). Combination rule preserved as an Observation per CLAUDE.md §13.5.
8. **Cutoff function f external** (Paper 32 `thm:cutoff_function_external` + `cor:cc_fine_tuning_extern` Sprint D5/D6): cutoff function f and its Mellin moments are external test-function data; no axiom in $\mathcal{A}$ selects moment values. Wald forces relations among moments; values are external. Cosmological-constant fine-tuning $\varphi(2)/\varphi(1)^2 \approx 10^{-124}$ formalised as external moment selection.

Plus one named structural-research target (Paper 32 `rem:chemistry_eta_analog`, Sprint Chemistry-Analog): the chemistry-side η-trivialization analog. Paper 18 `thm:eta_trivialization` forces inner-factor period ring to $M_1 \cup M_2$ via $\{\gamma_F, D_F\} = 0$; the chemistry-side analog would be a structural property of the FrozenCore $Z_{\text{eff}}(r)$ pipeline mirroring this chirality grading. Sprint-scale path unclear; substantial NCG-research target. Catalogue entries H6, H7 remain empirical-only pending.

The C-arc terminal state. Paper 32 `rem:c_arc_terminal_state` records the full corpus tally: eight theorem-grade non-selection results across four sectors (inner-factor / multi-focal-composition / α combination rule / gravity), one named substantial follow-on (chemistry η-trivialization analog), all subsidiary residuals (F4–F7 inherit from F1 + F4; I-entries via Gleason / E6 inheritance / open identification) accounted for.

The full forced/free seam is in a stable terminal state for the path-#1 vein C arc. Further sprints on the seam either tackle the named substantial chemistry η-trivialization analog or are residual subsidiaries.

6.4 The chemistry-side η-trivialization analog

A concrete sprint-scale follow-on remains. P5's failure modes match independent structural arguments on the SM side (η-trivialization, Paper 18 §IV.6) but not yet on the chemistry side. The chemistry-side analog would be a structural property of the FrozenCore $Z_{\text{eff}}(r)$ pipeline that mirrors the chirality-grading axiom $\{\gamma_F, D_F\} = 0$ on the SM side. Such a property would give a categorical reason for the chemistry-side inner-factor input-data entries (H6, H7) to be packing-unreachable. The candidate is well-defined but not currently exhibited; flagged as substantial NCG-research target in the W1e period-class memo [26].

6.5 The "second packing axiom" status

The Direction 2 NO-GO [24] and Read 2 NO-GO [25] establish that the current Paper 0 packing construction does not autonomously reach N_{gen} or the inner KO-dim. The named logical escape is a *sibling axiom* to Paper 0: a new primitive emitting fiber multiplicities, motivated by structural content the existing axiom system does not carry. For a sibling axiom to satisfy the standards of the present catalogue it would need to:

1. Be motivated by structural content the existing axiom system does not carry (not just postulate $N_{\text{gen}} = 3$ directly).
2. Have testable consequences beyond the multiplicity (e.g. predict CKM structure, generation mass hierarchy, or Yukawa–Higgs coupling pattern).
3. Be self-consistent with the Door 4 forced/free seam (i.e. not accidentally also force Yukawa values, which would contradict the seam theorem).

These requirements are well-defined but no current construction satisfies all three. The sibling-axiom direction remains the named long-range structural research target.

6.6 Higgs direction and the Hopf base

The Boyle–Farnsworth DGA parameterization [19] represents the Higgs by a unit vector $\hat{n} \in S^2$ specifying the direction of the $\mathbb{C}$ embedding in $\mathbb{H}$. The GeoVac Hopf base S^2 carries the master Mellin engine M_1 signature $\text{Vol}(S^2)/4 = \pi$. If $\hat{n}$ is identified with the Hopf-base S^2, this would provide a structural correlation between the Higgs orientation and the gauge-bundle base. The follow-on is flagged but not pursued at sprint scale; it sits at the boundary between Observation 3.1 family 2 (where the Higgs direction is currently catalogued) and the multi-focal-composition family.

6.7 Relationship to Connes–Marcolli calibration data in NCG-SM

The free / forced seam catalogued here is consistent with — and arguably a specific instance of — the more general phenomenon visible in the Connes–Marcolli noncommutative-geometric formulation of the Standard Model: the framework determines the gauge group, the algebra structure, and the spectral action machinery, but the Yukawa values and the generation count enter as external inputs. The GeoVac contribution to this conversation is twofold: the case-exhaustion-theorem-driven classification of which periods can and cannot appear on the inner factor (the $M_1 \cup M_2$ restriction via η-trivialization), and the empirical confirmation at 162 PSLQ cells that the SM Yukawas do not in fact land in this restricted period ring at low coefficient ceiling.

The natural follow-on at the cross-community level is engagement with the Brown / Marcolli / Glanois periods program. The two new structural theorems of Paper 18 §IV.6 (η-trivialization and tensor factorization) are paper-level handles for that engagement.

Honest scope

- **Observations-grade content.** This paper consolidates and frames; it does not prove a meta-theorem of the form "any observable on the GeoVac spectral triple is forced iff packing-reachable." Sprint C2 identifies packing-reachability as the 98.3%-accurate surface principle, but whether P5 graduates from observation to theorem is the open question of §6.1. The two-family structure of Observation 3.1 is empirical, supported by the discriminator-column patterns in the catalogue `docs/forced_free_seam.md`, and is preserved as the failure-mode decomposition beneath P5 rather than as evidence that two separate principles are needed.

- **Catalogue completeness.** The 60 entries (35 + 1 + 24) are the load-bearing inventory at the time of writing. We may have missed entries; the catalogue's structure permits straightforward addition of new rows as further sprints surface them. The cross-domain unity of §4 is supported by six independent instances surfaced over a year of internal work, which is past the point of coincidence but is not a closed enumeration.

- **Theorem-grade content used here, not produced.** The Forced-Count Theorem (Paper 32, `thm:forced_count`), the case-exhaustion theorem (Paper 32 §VIII), the master Mellin engine classification (Paper 18 §III.7), the η-trivialization theorem (Paper 18 §IV.6), Paper 22's angular-sparsity theorem, Paper 38's GH-convergence theorem, the Wald two-term-exactness $\Rightarrow$ pure Einstein argument, and the Door 4 series theorems are all cited; none is reproved here.

- **Empirical-only content used here.** The 162-cell Yukawa-PSLQ clean negative is empirical / numerical at coefficient ceiling $M \leq 10^3$; high-coefficient identities are not ruled out structurally. The W1e period-class diagnostic's 11-term sweep is similarly empirical. The cross-domain unity is six observed instances, not a theorem.

- **Named open follow-ons (deferred).** The chemistry-side η-trivialization analog (substantial NCG-research target); the sibling-axiom direction toward N_{gen} and inner KO-dim (long-range, no current handle); the Higgs-direction / Hopf-base identification (sprint-scale, NCG-reading-heavy, flagged).

Cross-references

Companion catalogue: `docs/forced_free_seam.md` (this writeup's companion data file, full 60-entry table).

Strengthens / consolidates: Paper 32 §VIII (Forced-Count Theorem, Sprint H1 verdict) [9]; Paper 18 §III.7 + §IV.6 (master Mellin engine + inner-factor input-data tier) [4]; Paper 35 (rest-mass vs. temporal-window projections) [11]; Paper 50 (F-theorem ladder, Door 1 forward-run) [16]; the prior catalogue `docs/forcing_catalogue.md` (2026-06-01).

Does not change: WH1 PROVEN status [13]; WH5 (α as projection constant); the Marcolli–van Suijlekom gauge-network lineage reading of Papers 25, 30, 32.

References

[1] J. Loutey, *Paper 0: Geometric Packing*, GeoVac papers/group3_foundations.

[2] J. Loutey, *Paper 7: The Dimensionless Vacuum*, GeoVac papers/group3_foundations.

[3] J. Loutey, *Paper 14: Qubit Encoding for the Composed Architecture*, GeoVac papers/group4_quantum_computing.

[4] J. Loutey, *Paper 18: Spectral-Geometric Exchange Constants*, GeoVac papers/group3_foundations.

[5] J. Loutey, *Paper 22: Potential-Independent Angular Sparsity*, GeoVac papers/group3_foundations.

[6] J. Loutey, *Paper 23: Nuclear Shell Model Qubit Hamiltonians and Fock Projection Rigidity*, GeoVac papers/group4_quantum_computing.

[7] J. Loutey, *Paper 24: The Bargmann–Segal Lattice for the 3D Harmonic Oscillator*, GeoVac papers/group3_foundations.

[8] J. Loutey, *Paper 28: QED on S^3*, GeoVac papers/group5_qed_gauge.

[9] J. Loutey, *Paper 32: The GeoVac Spectral Triple*, GeoVac papers/group1_operator_algebras.

[10] J. Loutey, *Paper 33: The 1+6+1 Partition of QED Selection Rules on S^3*, GeoVac papers/group5_qed_gauge.

[11] J. Loutey, *Paper 35: Time as Projection*, GeoVac papers/group6_precision_observations.

[12] J. Loutey, *Paper 36: Bound-State QED on Dirac-S^3*, GeoVac papers/group5_qed_gauge.

[13] J. Loutey, *Paper 38: State-space Gromov–Hausdorff Convergence of Truncated Camporesi–Higuchi Spectral Triples on S^3*, GeoVac papers/group1_operator_algebras.

[14] J. Loutey, *Paper 40: State-space Gromov–Hausdorff Convergence of Spectral Truncations of Compact Lie Groups with Bi-invariant Metric*, GeoVac papers/group1_operator_algebras.

[15] J. Loutey, *Paper 43: Lorentzian Extension at Finite Cutoff*, GeoVac papers/group1_operator_algebras.

[16] J. Loutey, *Paper 50: Boundary CFT_3 Observables on the Truncated GeoVac Spectral Triple*, GeoVac papers/group1_operator_algebras.

[17] J. Loutey, *Paper 51: Gravity from the GeoVac Spectral Action*, GeoVac papers/group5_qed_gauge.

[18] J. Bertrand, *Théorème relatif au mouvement d'un point attiré vers un centre fixe*, C. R. Acad. Sci. Paris **77** (1873) 849–853.

[19] L. Boyle and S. Farnsworth, *A new algebraic structure in the standard model of particle physics*, arXiv:1604.00847.

[20] A. Döring and J. Harding, *Abelian subalgebras and the Jordan structure of a von Neumann algebra*, Houston J. Math. **42** (2016) 559–568; arXiv:1009.4945. (Thm 3.4: an order isomorphism of the poset of abelian subalgebras is induced by a Jordan *-isomorphism for algebras with no type I_2 summand and $\neq \mathbb{C} \oplus \mathbb{C}$.)

[21] J. Hamhalter, *Isomorphisms of ordered structures of abelian C*-subalgebras of C*-algebras*, J. Math. Anal. Appl. **383** (2011) 391–399; doi:10.1016/j.jmaa.2011.05.035. (C*-algebra extension of the reconstruction theorem; uniqueness holds as long as A is not two-dimensional.)

[22] A. J. Lindenhovius, *Classifying finite-dimensional C*-algebras by posets of their commutative C*-subalgebras*, Int. J. Theor. Phys. **54** (2015) 4615–4635; arXiv:1501.03030. (Thm 1: for finite-dimensional A, $\mathcal{C}(A)$ determines A up to *-isomorphism, with no type I_2 exception.)

[23] GeoVac sprint memo: `debug/sprint_yukawa_pslq_memo.md` (2026-06-03): Sprint Yukawa-PSLQ clean negative across 162 PSLQ cells.

[24] GeoVac sprint memo: `debug/seam_packing_scoping_memo.md` (2026-06-03): Direction 2 NO-GO.

[25] GeoVac sprint memo: `debug/sprint_read2_n_gen_scoping_memo.md` (2026-06-03): Read 2 NO-GO on Hopf-tower-to-representation shortcut.

[26] GeoVac sprint memo: `debug/sprint_w1e_period_class_memo.md` (2026-06-04): W1e period-class diagnostic, zero genuine outer-factor identifications.

[27] GeoVac sprint memo: `debug/sprint_c2_principle_hunt_memo.md` (2026-06-08): Sprint C2 principle hunt, packing-reachability at 98.3% accuracy on the 60-entry catalogue.

[28] GeoVac sprint memo: `debug/sprint_c3_n_gen_non_selection_memo.md` (2026-06-08): Sprint C3 N_{gen} non-selection theorem; theorem-grade upgrade of Direction 2 + Read 2 NO-GO scopings.

[29] GeoVac sprint memo: `debug/sprint_f3_ko_dim_non_selection_memo.md` (2026-06-08): Sprint F3 inner KO-dim non-selection theorem; theorem-grade upgrade via packing-is-kinematic + KO-dim-is-real-structure-data composition.

[30] GeoVac sprint memo: `debug/sprint_e7_e8_single_cutoff_memo.md` (2026-06-08): Sprint E7/E8 single-cutoff spectral action theorem; closes multi-loop QED renormalization sub-sector of multi-focal-composition wall (E7, E8, G3, G4).

[31] GeoVac sprint memo: `debug/sprint_g1_g2_g5_spatial_composition_memo.md` (2026-06-08): Sprint G1/G2/G5 spatial-composition radial wall theorem; closes spatial-composition sub-sector of multi-focal-composition wall (G1, G2, G5). Multi-focal-composition wall now fully theorem-bound at the canonical-rep level under two structurally distinct theorems.

[32] GeoVac sprint memo: `debug/sprint_c_arc_closure_e6_d5d6_chemistry_memo.md` (2026-06-08): Sprint C-arc closure; E6 (no-single-mechanism K) + D5/D6 (cutoff function external) theorems + chemistry-side η-trivialization analog scoping; C-arc terminal-state remark tally of eight theorem-grade non-selection results.